

# Attention Helps Small Models

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## Abstract

We study how self-attention enables small models to read long inputs with a feed-forward network. Results are strong.

## 1 Introduction

Recurrent networks process tokens one at a time, so training is slow. Models that rely on attention read all tokens at once and train much faster than recurrent ones.

A second paragraph begins here and talks about parallel training. It also says it was trained for 3.5 days on 8 GPUs with 2 nodes and no extra data and mentions that the cost of a step grows with the square of the input length.  
Cost: 3.5 days, 8 GPUs, 2 nodes.

2 Method

The model has an encoder and a decoder. Each layer has two sub-layers: attention and a feed-forward network, joined by residual connections that carry the signal around each sub-layer and make deep stacks easy to train with a sin-

gle learning rate. Training used eight GPUs.  
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2.1 Training

Table 1 lists the results of the experiments.

| Model | Params | BLEU | Cost |
|-------|--------|------|------|
| Base  | 65     | 27.3 | 3.3  |
| Big   | 213    | 28.4 | 2.3  |

### 3 Conclusion

Attention helps small models read long inputs, and it does so with less training cost than recurrent models.

Results are good.

Costs are low.

Code is open.



Figure 1: The attention layer

More results are listed in the table of the appendix and in the next

Figure 3: Costs of training

#### References

[1] A. Vaswani et al. Attention is all you need. 2017.